

Exp-1: OC, SC and Load Tests on 1- ϕ Transformer

1. Aim of this Experiment:

The objective of this experiment is to determine the values of R , X , B_m and G_i through short circuit and open circuit test as well as to verify the equivalent circuit representation through a load test.

2. Circuit Description and Test Arrangement:

The equivalent circuit of a 1- ϕ transformer in approximate form is shown in Fig 1. Here, R and X are the total resistance and leakage reactance, respectively of the transformer windings referred to a particular side. Similarly, B_m and G_i are the magnetizing susceptance and the iron loss conductance, respectively, referred to a particular side.

The circuit arrangements for open circuit test, short circuit test, and load test are shown in Fig 2, 3, 4 respectively.

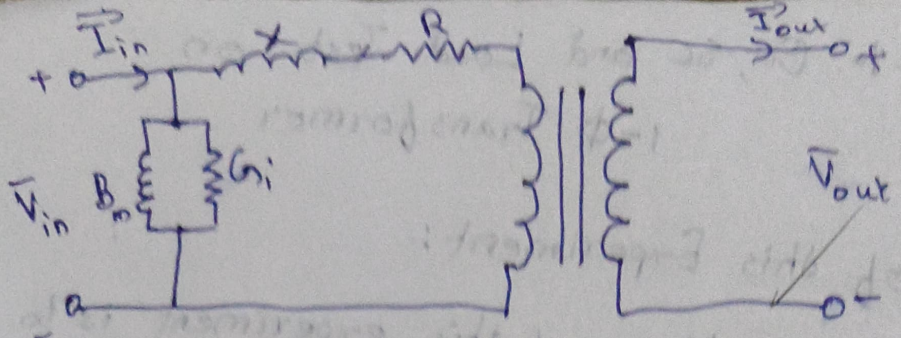


Fig 1- Equivalent circuit representation of a 1- ϕ transformer.

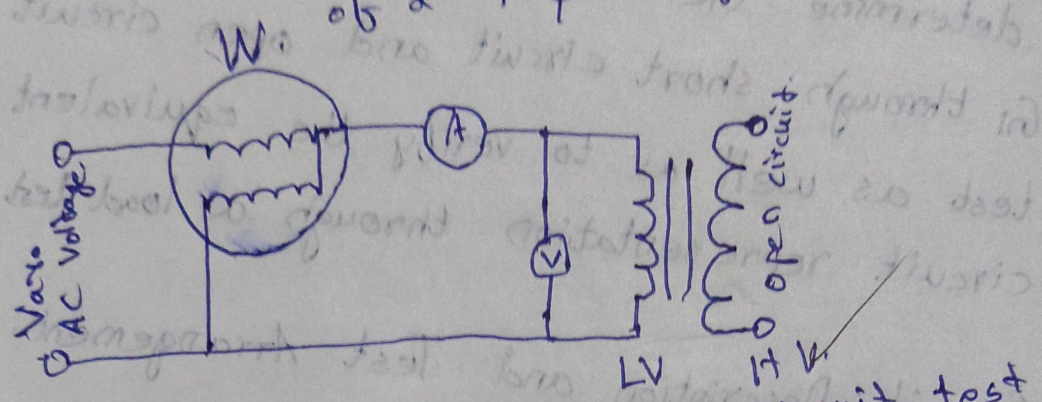


Fig-2 1- ϕ transformer open circuit test arrangement.

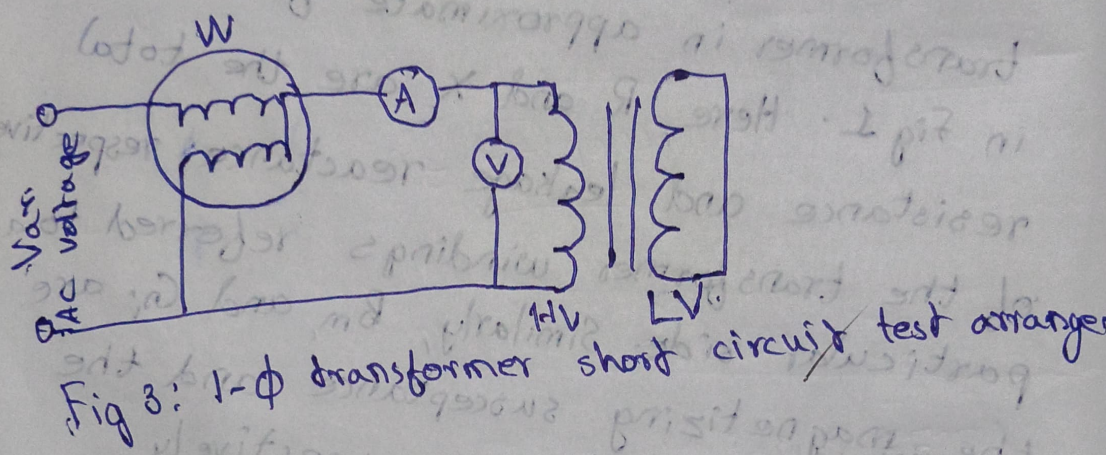


Fig 3: 1- ϕ transformer short circuit test arrangement.

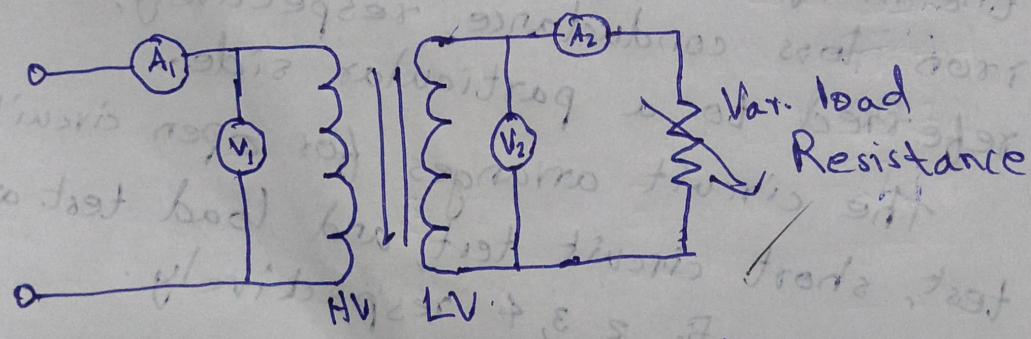


Fig 4. 1- ϕ transformer load test arrangement.

3. Test Procedure

i) OPEN CIRCUIT TEST:

a) Apply rated voltage on the LV side.

b) Note down the voltmeter, ammeter, and wattmeter readings (indicated by V_{oc} , I_{oc} , P_{oc}).

c) Determine G_i and B_m by using the following formulae.

$$G_i = \frac{P_{oc}}{V_{oc}^2} \quad Y = \frac{I_{oc}}{V_{oc}} \quad B_m = \sqrt{Y^2 - G_i^2}$$

ii) SHORT CIRCUIT TEST:

a) Establish rated current on the HV side by adjusting the input voltage.

b) Note down the voltmeter, ammeter, and wattmeter readings (indicated by V_{sc} , I_{sc} and P_{sc}).

c) Determine R and X by using the following formulae.

$$R = \frac{P_{sc}}{I_{sc}^2} \quad Z = \frac{V_{sc}}{I_{sc}} \quad X_s = \sqrt{Z^2 - R^2}$$

iii) LOAD TEST:

a) Apply rated voltage on HV side.

b) Take the voltmeter and ammeter readings (indicated by V_{L1} , V_{L2} , I_{L1} , I_{L2}) for different settings of the resistance bank.

c) For a particular setting of the resistance bank, the value of load resistance (indicated by R_L) can be found from the readings of Voltmeter 2 and Ammeter 2.

d) For a given value of load resistance and input voltage, calculate value of I_{L1} from equivalent circuit.

4. Results and Discussions

OPEN CIRCUIT TEST:

$$V_{oc} = \underline{\underline{120}}$$

$$G_i \text{ (referred to LV side)} = \frac{9.722 \times 10^{-4} \Omega}{1.655^2} = 9.722 \times 10^{-4} \Omega \text{ (referred to LV side)}$$

$$I_{oc} = \underline{\underline{0.23}}$$

$$G_i \text{ (referred to HV side)} = \frac{2.86 \times 10^{-2} \Omega}{4.59 \times 10^4 \Omega} = 2.86 \times 10^{-2} \Omega \text{ (referred to HV side)}$$

$$P_{oc} = \underline{\underline{14W}}$$

SHORT CIRCUIT TEST:

$$V_{sc} = \underline{\underline{10V}}$$

$$R \text{ (referred to HV side)} = \frac{2.96 \Omega}{3.96 \Omega} = 2.96 \Omega \text{ (referred to HV side)}$$

$$I_{sc} = \underline{\underline{2.09A}}$$

$$P_{sc} = 12W$$

$$I_{sc} = \underline{\underline{4.19A}}$$

$$R \text{ (referred to LV side)} = \frac{0.5778 \Omega}{1.3215 \Omega} = 0.5778 \Omega \text{ (referred to LV side)}$$

$$V_{sc} = \underline{\underline{6V}}$$

$$P_{sc} = 10W$$

LOAD TEST:

V_{L1} (measured)	V_{L2} (measured)	I_{L1} (measured)	I_{L2} (measured)	R_L (calculated)	I_H (calculated)
240V	126V	0.277A	0.3A		
240V	125	0.524A	0.51A	154.32Ω	0.5525A
237.2	123	0.761A	1.31A	93.89Ω	0.7744A
237.4	122	1.046A	1.8A	67.77Ω	1.0083A
235.8	121	1.313A	2.28A	53.07Ω	1.2214A
236.1	121	1.545A	2.70A	44.81Ω	1.2765A
					1.414A

5. Conclusions

Results

OPEN CIRCUITS

Referred to ~~FOR~~ LV side

$$V_{oc} = 120V \quad G_i = 9.722 \times 10^{-4} \Omega^{-1}$$

$$I_{oc} = 0.23A \quad P_{oc} = 14W \quad B_m = 1.65 \times 10^{-3} \Omega^{-1}$$

Referred to HV side

$$V_{oc} = 240V \quad G_i = 2.864 \times 10^{-4} \Omega^{-1}$$

$$I_{oc} = 0.13A \quad P_{oc} = 16.5W \quad B_m = 4.59 \times 10^{-4} \Omega^{-1}$$

SHORT CIRCUITS

$$\text{Rated } I_{sc} = 2.04A \quad V_{sc} = 10V \quad Z = 4.9\Omega$$

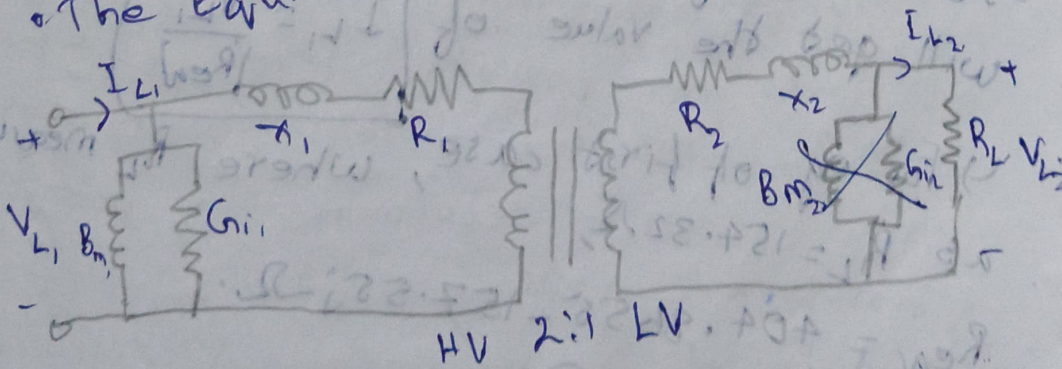
$$P_{sc} = 12W \quad R = 2.48\Omega$$

$$X_{HV} = 3.96 \Omega$$

LOAD TEST

R, X calculated using the results from short circuit and open circuit.

The Equivalent circuit looks like

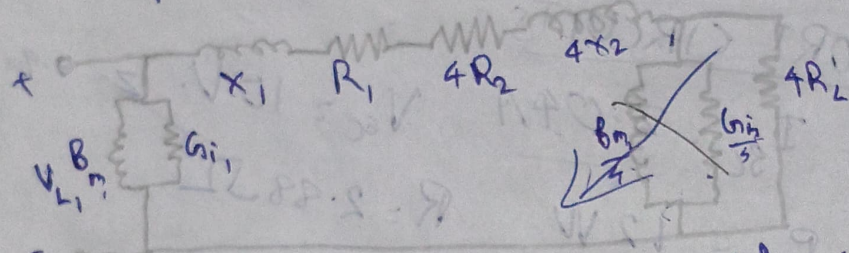


• We convert everything to base units by dividing by base values on both sides so that we can treat it as a single circuit without transformer.

• $V_{L1} \text{ base} = \frac{1}{2}$, $B_{m1} \text{ base} = 1$, $G_{i1} \text{ base} = 1$
 $X_{L1} \text{ base} = 1$, $R_{L1} \text{ base} = 1$, $I_{L1} \text{ base} = 1$.

• $V_{L2} \text{ base} = \frac{1}{2}$, $B_{m2} \text{ base} = 4$, $G_{i2} \text{ base} = 4$
 $X_{L2} \text{ base} = \frac{1}{4}$, $R_{L2} \text{ base} = \frac{1}{4}$, $I_{L2} \text{ base} = 2$.

• $\therefore$ Eq. circuit looks like,



• (Circuit is with ref. of HV side, so all impedances on LV side are multiplied by 4)

$$R_{eq} = \left[\left(\left(\frac{1}{B_{m1}} \right) \cdot 4 \right) \parallel \left(\left(\frac{1}{G_{i1}} \right) \cdot 4 \right) \parallel \left(\left(\frac{1}{4R_{L2}} \right) \cdot 4 \right) \right] + 4X_{L2} + 4R_{L2} + R_{L1} + X_{L1}$$

$\Rightarrow$ Substituting values in this eqn, we will get the value of $I_{L1} = \frac{V_{L1}}{R_{eq}}$

$\hookrightarrow$ For eg, of first case, where R_{L2} means as $R_{L2} = 154.32 \Omega$,

$$R_{eq} = 404.8655 - 157.55j \Omega$$

$$\therefore R_{eq} = 434.36 \Omega$$

$$\therefore V_{L1} \text{ for first} = 240V$$

$$I_{L1} = \frac{V_{L1}}{R_{eq}} = \frac{240}{436.36} = \underline{\underline{0.5525 A}}$$

Similarly, we can measure I_{L1} for other tests by changing values.

Conclusion:

- OC test gave shunt parameters, B_m & b_{11} .
- SC test gave series parameters R & X , which together define eq. circuit of transformer.
- Calculated & measured I_{L1} values were in 5.5 - 9.26% tolerance error with each other.

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Exp-3

Speed Control of DC Shunt Motor

1. Objective

The objective of the experiment is to be familiar with the 'armature control' and 'field control' mechanisms of the DC shunt motor.

2. Description

For a given input voltage, the speed of a motor falls down when the load on the motor is increased. In the same way, the motor speed increases when the load on the motor is reduced. However, for any practical purpose it is intended to run the motor at constant speed for all the loading conditions. $\therefore$ it is necessary to implement some speed control mechanism so as to bring the motor speed back to the original value after any load variation.

The following scenarios give the applications of DC motors.

- ✓ Where the speed is required to remain almost constant from no-load to full-load.
- ✓ Where the load has to be driven at a number of speeds and any one of which is nearly constant.

Various applications of DC shunt motor include lathe machines, centrifugal pumps, fans, blowers, conveyors, lifts, weaving machines, rolling mills, spinning machines, etc.

The speed control for a DC shunt motor can be effectively carried out by suitably adjusting the armature and field resistance based on the following relations.

$$E_b = \frac{N P \Phi Z}{60 A}$$

$$\Rightarrow N = \frac{E_b \cdot 60 A}{P \Phi Z} = \left(\frac{E_b}{\Phi} \right) \left(\frac{60 A}{P Z} \right)$$

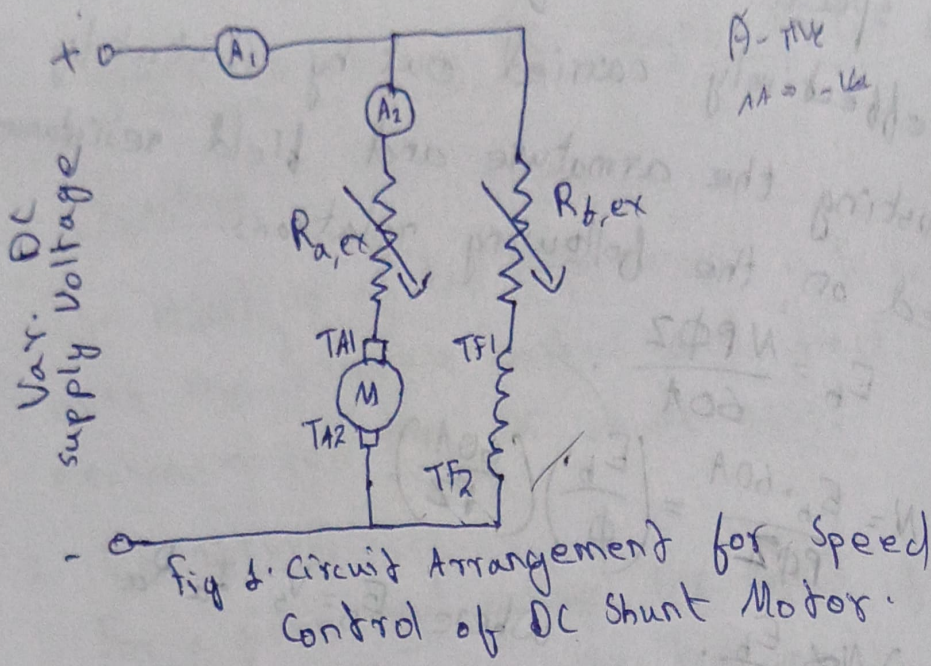
$$\Rightarrow N \propto \frac{E_b}{\Phi} \quad \text{Where } E_b = V_s - I_a R_a$$

$$\therefore N \propto \frac{V_s - I_a R_a}{\Phi}$$

(Where E_b = back emf, N = speed in rpm, P = no. of pole, Φ = flux/pole, Z = no. of armature conductors, A = parallel paths, V_s = supply voltage, I_a = armature current, R_a = armature resistance).

3. Circuit arrangement

The circuit arrangement for the experiment "speed control of DC shunt motor" is shown in Fig 1. Here, TA1, TA2 are terminals of armature winding and TF1, TF2 are the terminals of field winding. Resistances $R_{a,ex}$ and $R_{f,ex}$ are the external resistances connected in series with the armature winding and field winding respectively.



4. Procedure Speed Control

1. Before starting exp., note down readings of armature and field resistances.
2. Now, connect circuit diagram as per Fig 1.
3. Set armature rheostat at max. position and field rheostat at min. position.
4. Slowly increase supply voltage up to 150V and keep it const. at this value.
5. Try to bring machine speed to 1500 rpm by first adjusting the armature resistance and then by adjusting the field resistance.
6. Note down values of armature / field resistances & currents when the machine speed is 1500 rpm.
7. Repeat same procedure for i/p voltages 190V, 200V, 210V and 220V. Ensure that motor speed and armature current do not exceed rated values, which are 1500 rpm and 4.5A.

Speed Regulation

8. Vary the field rheostat for diff. values and observe occurrence of above rated speed values.
9. Similarly observe below rated speed values by varying armature rheostat.

5. Results

I. Under Rest Condition:

$$\text{field Resistance } (R_f) = 208.4 \Omega$$

$$\text{Armature Resistance } (R_a) = 3.4 \Omega$$

II Speed Control Test Observations:

Vp Voltage	Corresponding speed	Total armature resistance (self + inject) to 1500 rpm	Total field resistance (self + inject) to 1500 rpm	Armature Current (I_a)	Field Current (I_f)
170V	1508	3.4	439.02 Ω	1.42 A	0.41
190V	1502	3.4	395.83 Ω	1.76 A	0.48
200	1505	3.4	363.63 Ω	1.65 A	0.55
210	1498	3.4 Ω	362.06 Ω	1.61 A	0.58
220	1503	3.4 Ω	328.52 Ω	1.53 A	0.68

306.4+

208.3

III Speed Regulation Test Observations:

Above Rated Speed Measurements:

i/p voltage	Speed	Total field resistance (self + injected)	Field Current (I_f)
220V	1520	349.206	0.63
	1550	349.310	0.58
	1585	407.407	0.54
	1625	440	0.50
	1665	468.085	0.47

Below Rated Speed Measurements:

i/p Voltage	Speed	Total armature resistance (self + injected)	Armature Current (I_a)
220V	1455	141.025 7.72	1.56 A
	1400	142.845 12.9	1.54 A
	1350	143.792 17.7	1.53 A
	1300	144.731 22.5	1.52 A
	1250	145.695 27.4	1.51 A
	1250		

6. Conclusions

i) Speed Control v/s Speed Regulation

- By deliberately adjusting R_a & R_f , motor speed was brought to rated 1500 rpm for every supply (150V-220V).

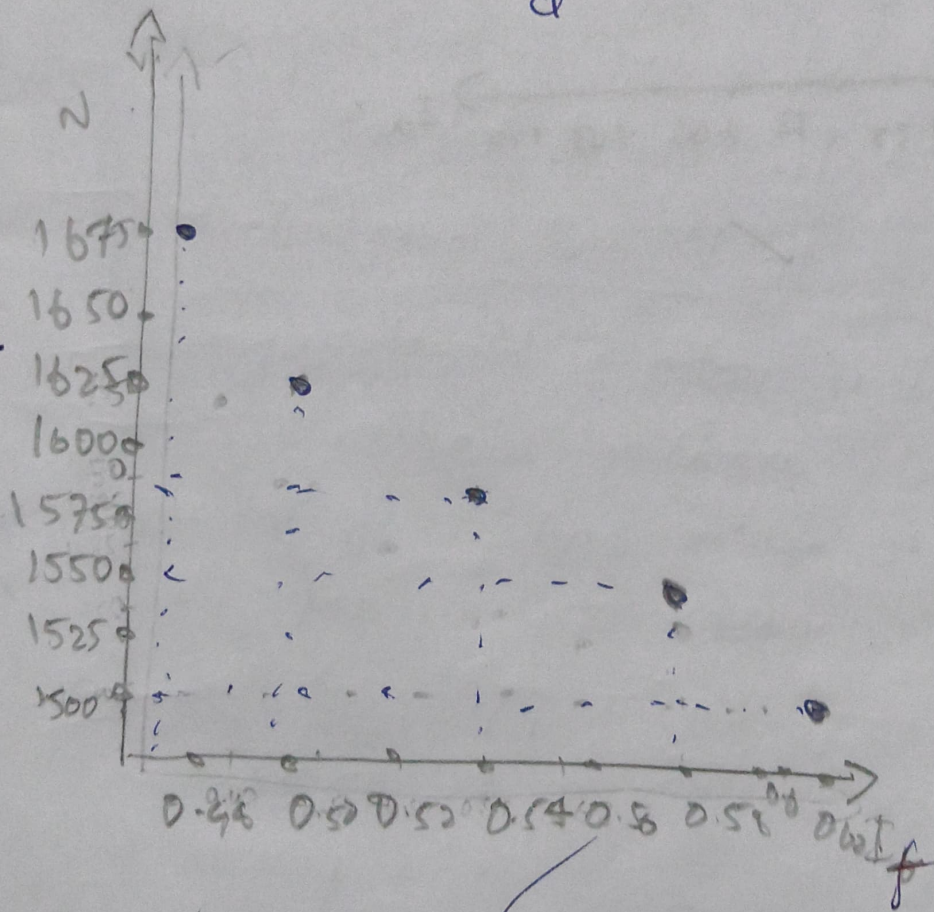
As V_s & I_f was decreasing in order to $T^s I_f$ & thus ϕ which leads to $(V_s - I_a R_a)$ remains const.

→ For speed regulation, V_s is fixed & excitation is varied & change in rpm is observed.

ii) Armature control $(R_a)_{ex} \uparrow^s$ and $I_a(R_a)_{ex} \uparrow^s \therefore V_s - I_a(R_a)_{ex} \downarrow^s$ which leads to $N \downarrow^s$.

• Field control: $(R_f)_{ex} \uparrow^s$ which leads to $I_f \downarrow^s$, as $\phi \propto I_f \Rightarrow N \downarrow^s$.

$$\therefore N = k \frac{(V - I_a R_a)}{\phi}$$



Exp - 4

V and Inverted V Curves of Synchronous Motor

1. Objective

The objective of this experiment is to study the following characteristics of a synchronous motor.

- Variation of armature current against variation in field current.
- Variation of power factor against variation in field current.

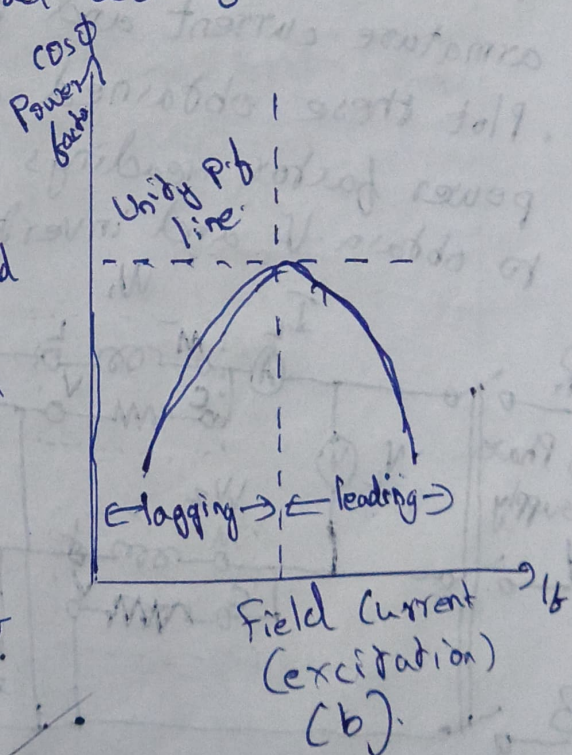
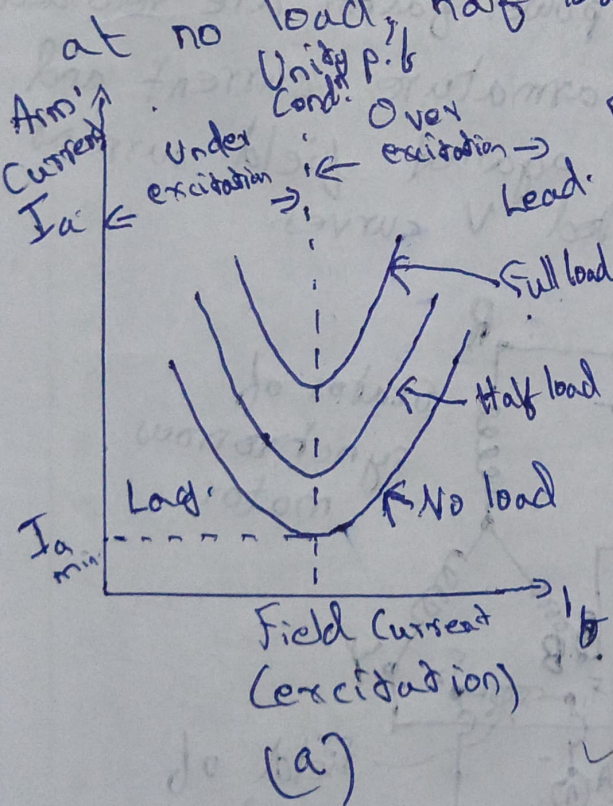
2. Introduction to V and Inverted V Curves

A synchronous motor is doubly excited machine, its armature winding is energized from an AC source and its field winding from the DC source. Total air gap flux is the resultant of 2 fluxes produced by the AC and DC excitation. The DC excitation which operates the motor at unity power factor is called the nominal or normal excitation.

If the field current is made less than normal excitation (under excitation), then the deficiency in air gap flux is made up by the armature MMF. So the armature (stator) winding draws a magnetizing current or lagging VA from the AC source and as a

result motor operates in lagging power factor. Similarly, if the field current is made more than the normal excitation (over excited) motor operates in leading power factor.

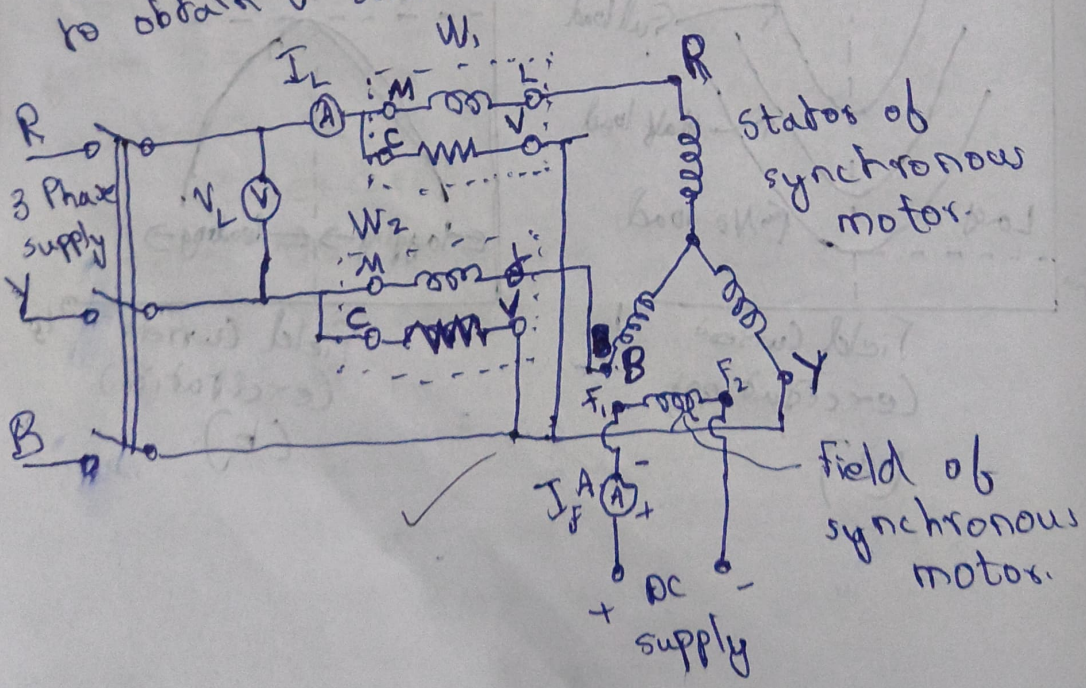
If we draw the variation of armature current and power factor vs field current the curves appear as V and inverted V respectively. The feature of synchronous motor that operates in leading power factor when over excited is utilized in power factor correction applications. Synchronous machines have paraboliz type characteristics. The following figure shows the variation of armature current and power factor with field current at no load, half load and full load conditions.



3. Circuit Arrangement & Experiment

The circuit arrangement for conducting the experiment is shown as follows, where the armature current is measured by the line ammeter and power factor by Two-Wattmeter method. The field current is measured by the DC ammeter in the field circuit.

- Connections are made as shown in circuit diagram.
- Variable 3 phase supply is increased gradually to rated value, and then field current is increased till unity power factor is observed.
- Now varying the field current above and below the nominal excitation corresponding armature current and power factor are recorded.
- Plot these obtained armature current and power factor readings against field current to obtain V and inverted V curves.



4. Results.

V Curve: I_f vs I_a

S.No	Field current (I_f) (A)	Armature Current (I_a) (A)
1	0.22	3.96
2	0.30	3.26
3	0.27	3.50
4	0.33	3.05
5	0.36	2.75
6	0.45	2.30
7	0.53	1.66
8	0.60	0.95
9	0.70	0.50
10	0.83	1.50
11	0.93	3.0
12	0.97	5

Inverted V Curve: I_f vs $\cos \phi$

S.No	Field Current (I_f)	W_1	W_2	Power factor ($\cos \phi$)
1	0.22	1150	-660	0.154
2	0.30	995	-548	0.165
3	0.27	1058	-591	0.161
4	0.33	948	-510	0.171
5	0.36	880	-461	0.178
6	0.45	732	-333	0.211
7	0.53	568	-200	0.267
8	0.60	470	-83	0.418
9	0.70	211	123	0.910
10	0.83	190	280	0.944
11	0.93	85	360	0.693
12	0.97	50	430	0.589

$$\cos \phi = \cos \left(\tan^{-1} \left[\frac{\sqrt{3}(W_1 - W_2)}{W_1 + W_2} \right] \right)$$

5. Conclusion

- Before peak occurs in the inverted V characteristics, I_f is lagging so, motor behaves like an inductor as in inductor voltage leads current. 111^{th} after peak, motor behaves like cap bcz current leads.
- Through observation, before range of $0.7-0.43 A$ I_f , it is inductor and higher, it is capacitor.
- Operation of syn condenser is done in over excited region. i.e., when I_f is leading. This is done so that machine delivers reactive power to supply. It run unloaded. Mechanically it consumes very less real power while generating leading VARs which is, what we want and after, we see it crosses $0.7-0.43 A$ region.

= Applications

- Power factor correction:
Used alongside induction motors to elevate PF towards unity.
- Transmission line voltage regulation;
Used to stabilize line voltage by absorbing or injecting reactive VARs.

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